Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A plano-convex lens acts like a concave mirror of $28\ \text{cm}$ focal length when its plane surface is silvered and like a concave mirror of $10\ \text{cm}$ focal length when its curved surface is silvered. What is the refractive index of the material of the lens?

(A) 1.50

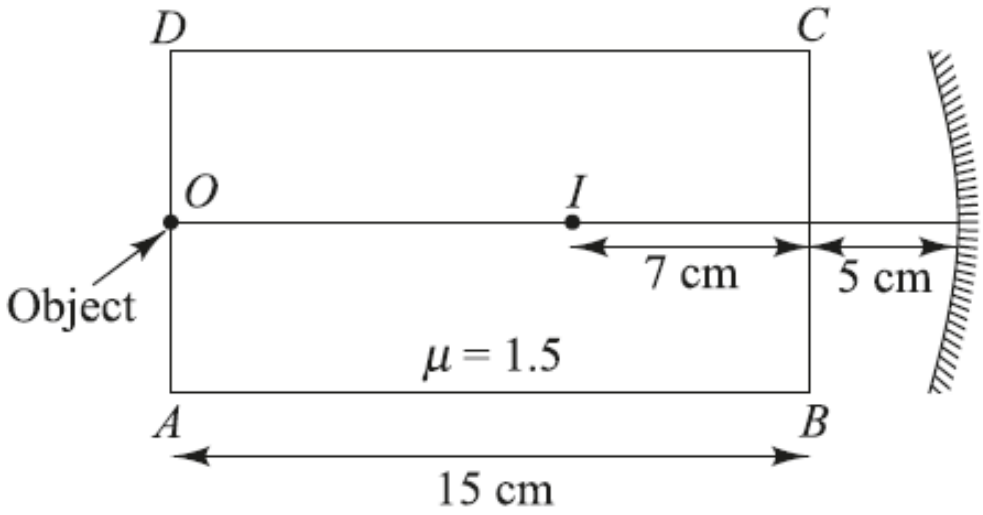
(B) 1.55

(C) 1.60

(D) 1.65

Q.2

A glass slab $ABCD$ of refractive index $\mu = 1.5$ and length $AB = 15\ \text{cm}$ has a concave mirror fitted at a distance of $5\ \text{cm}$ from side BC as shown in Fig. 16.60. A luminous object O is fixed on side AD and its image I is formed at distance of $7\ \text{cm}$ from side BC as shown. The focal length of the mirror is



(A) $\frac{15}{4}\ \text{cm}$

(B) $\frac{18}{5}\ \text{cm}$

(C) $\frac{20}{3}\ \text{cm}$

(D) $\frac{25}{6}\ \text{cm}$

Q.3

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A car is fitted with a convex side-view mirror of focal length $20\ \text{cm}$. A second car $2.8\ \text{m}$ behind the first car is overtaking the first car at a relative speed of $15\ \text{m/s}$. The speed (inm/s) of the image of the second car as seen in the mirror of the first one is :

(A) $\frac{1}{15}\ \text{m/s}$

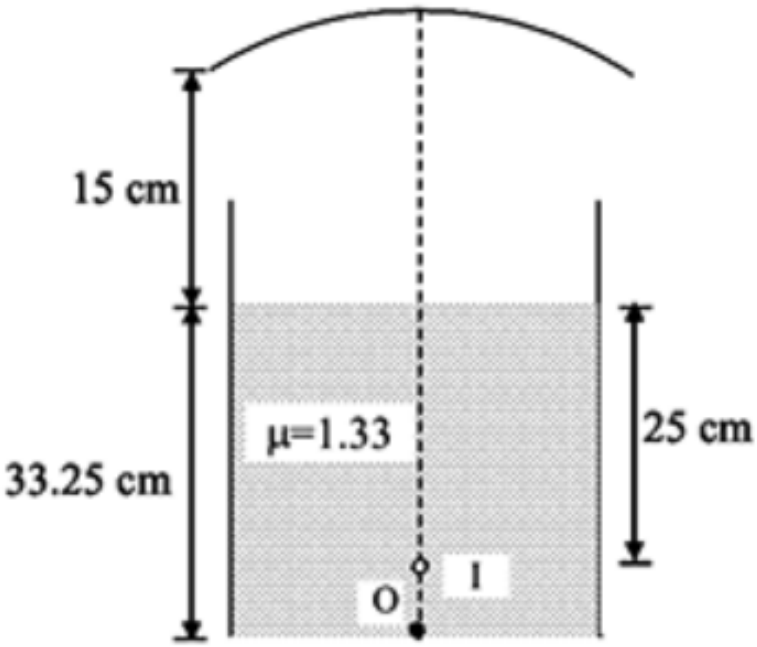
(B) $10\ \text{m/s}$

(C) $15\ \text{m/s}$

(D) $\frac{1}{10}\ \text{m/s}$

Q.4

A container is filled with water $(\mu = 1.33)$ upto a height of $33.25\ \text{cm}$. A concave mirror is placed 15 cm above the water level and the image of an object placed at the bottom is formed $25\ \text{cm}$ below the water level. Focal length of the mirror is



(A) $15\ \text{cm}$

(B) $20\ \text{cm}$

(C) $-18.31\ \text{cm}$

(D) $10\ \text{cm}$

Q.5

Two mirrors, one concave and the other convex, are placed $60\ \text{cm}$ apart with their reflecting surfaces facing each other. An object is placed $30\ \text{cm}$ from the pole of either of them on their axis. If the focal lengths of both the mirrors are $15\ \text{cm}$, the position of the image formed by reflection, first at the convex and then at the concave mirror, is:

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(A) $19.09\ \text{cm}$ from the pole of the concave mirror

(B) $19.09\ \text{cm}$ from the pole of the convex mirror

(C) $11.09\ \text{cm}$ from the pole of the concave mirror

(D) $11.09\ \text{cm}$ from the pole of the convex mirror

Q.6

I is the image of a point object O formed by spherical mirror, then which of the following statement is incorrect? (Take real or virtual objects at finite distances from pole)

(A) If O and I are on the same side of the principal axis, then they have to be on opposite sides of the mirror

(B) If O and I are on opposite sides of the principal axis, then they have to be on same side of the mirror

(C) If O and I are on opposite sides of the principal axis, then they have to be on opposite sides of the mirror as well

(D) If O is on principal axis, then I has to lie on principal axis only

Q.7

An object is moving with speed v_0 towards a spherical mirror with radius of curvature R, along the central axis of the mirror. The speed of the image with respect to the mirror is (U is the distance of the object from mirror at any given time t)

(A) $+\left(\frac{R}{U-2R}\right)v_0^2$

(B) $-\left(\frac{R}{R-2U}\right)^2 v_0$

(C) $-\left(\frac{R}{2U-2R}\right)^2 v_0$

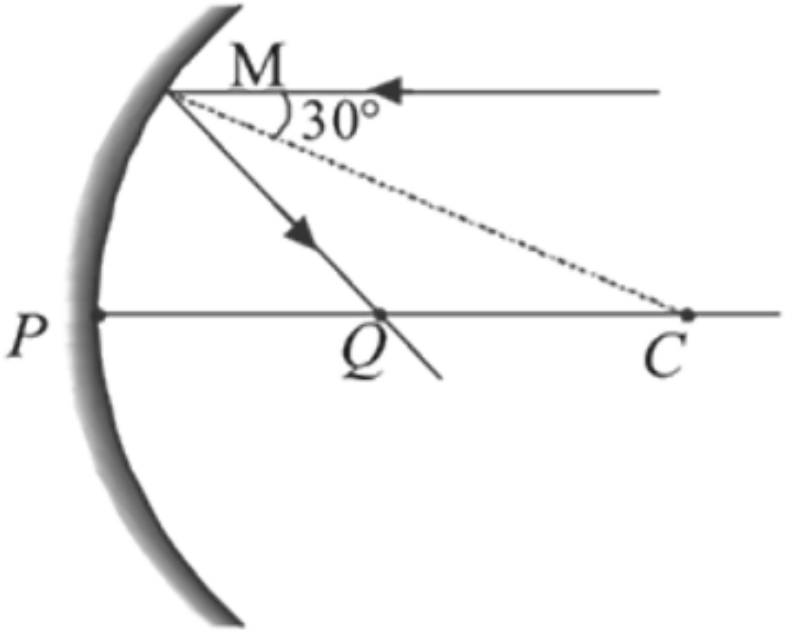
(D) $+\left(\frac{R}{2U-2}\right)v_0^2$

Q.8

A ray parallel to principal axis is incident at 30° from normal on concave mirror having radius of

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curvature R. The point on principal axis where rays are focussed is Q such that PQ is



(A) $\frac{R}{2}$

(B) $\frac{R}{\sqrt{3}}$

(C) $\frac{2\sqrt{R}-R}{\sqrt{2}}$

(D) $R\left(1-\frac{1}{\sqrt{3}}\right)$

Q.9

A concave mirror is placed on a horizontal table, with its axis directed vertically upwards. Let O be the pole of the mirror and C its centre of curvature. A point object is placed at C. It has a real image, also located at C. If the mirror is now filled with water, the image will be.

(A) real, and will remain at C.

(B) real, and located at a point between C and ∞.

(C) virtual, and located at a point between C and O.

(D) real, and located at a point between C and O

Q.10

A point object P moves towards a stationary convex mirror with a constant speed v, along the optical axis. The speed of the image

(A) is always less than v

(B) may be greater than, equal to or less than v, depending upon the position of P

(C) is always greater than v

(D) None of these

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Q.11

A short linear object of length ℓ lies along the axis of a concave mirror of focal length f at a distance u from the pole of the mirror. The size of the image is approximately equal to

(A) $\ell\left(\frac{u-f}{f}\right)^{1/2}$

(B) $\ell\left(\frac{u-f}{f}\right)^{2}$

(C) $\ell\left(\frac{f}{u-f}\right)^{1/2}$

(D) $\ell\left(\frac{f}{u-f}\right)^{2}$

Q.12

A point object ' O ' is at the centre of curvature of a concave mirror. The mirror starts to move at a speed u, in a direction perpendicular to the principal axis. Then, the initial velocity of the image is

(A) $2\mathrm{u}$, in the direction opposite to that of mirror's velocity

(B) $2\mathrm{u}$, in the direction same as that of mirror's velocity

(C) zero

(D) u, in the direction same as that of mirror's velocity

Q.13

You are asked to design a shaving mirror assuming that a person keeps it $10\ \mathrm{cm}$ from his face and views the magnified image of the face at the closest comfortable distance of $25\ \mathrm{cm}$. The radius of curvature of the mirror would then be :

(A) $60\ \mathrm{cm}$

(B) $-24\ \mathrm{cm}$

(C) $-60\ \mathrm{cm}$

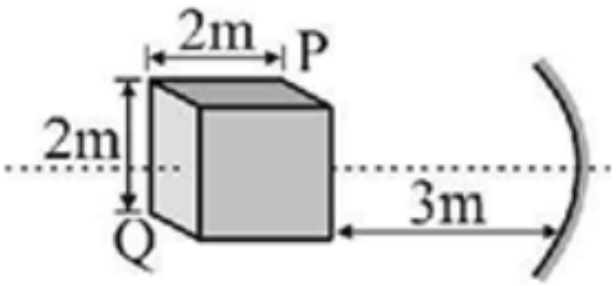
(D) $24\ \mathrm{cm}$

Q.14

A cube of side $2\ \mathrm{m}$ is placed in front of a concave mirror of focal length $1\ \mathrm{m}$ with its face P at a distance of $3\ \mathrm{m}$ and face Q at a distance of $5\ \mathrm{m}$ from the mirror. The distance between the image of face

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P and Q is



(A) $1\ \mathrm{m}$

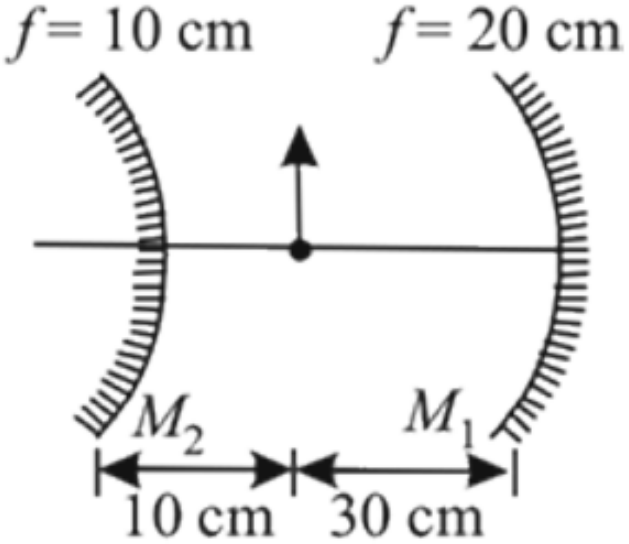
(B) $0.5\ \mathrm{m}$

(C) $0.75\ \mathrm{m}$

(D) $0.25\ \mathrm{m}$

Q.15

In figure find the total magnification after two successive reflections first on M_1 and then on M_2.



(A) $+1$

(B) -2

(C) $+2$

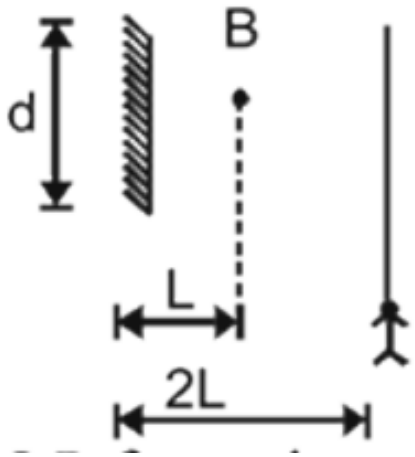
(D) -1

Q.16

A point source of light B is placed at a distance L in front of the centre of a mirror of width ' d hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance $2L$ from it as shown in fig. The greatest distance over which he can see the image of the light source in the

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mirror is



(A) $d/2$

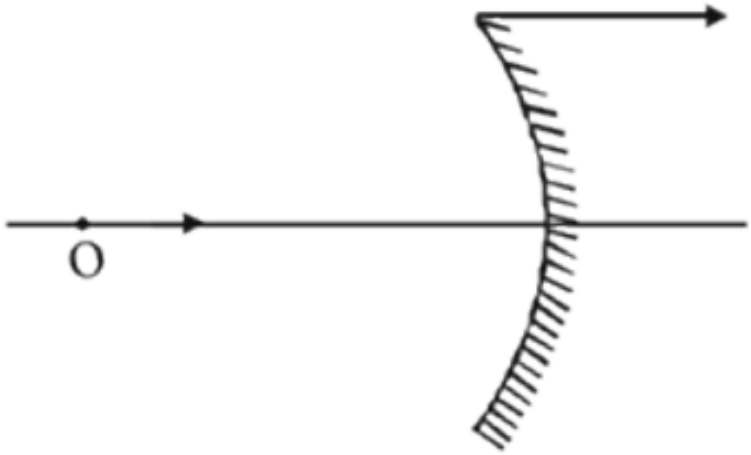
(B) d

(C) $2d$

(D) $3d$

Q.17

An object kept on the principle axis is moving in the same directions as that of mirror as shown in figure. Speed of object and mirror is $10\ \mathrm{m/s}$ and $40/\ 13\ \mathrm{m/s}$. Radius of the curvature of the mirror is 20 cm. What should be the distance of object from the mirror at this instant so that the image is stationary?



(A) $25\ \mathrm{cm}$

(B) $45\ \mathrm{cm}$

(C) $37.5\ \mathrm{cm}$

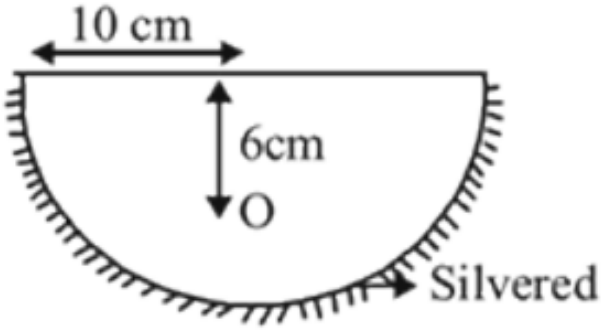
(D) $15\ \mathrm{cm}$

Q.18

A hemispherical glass body, $10\ \mathrm{cm}$ of radius $10\ \mathrm{cm}$ and refractive index 1.5 is silvered on its curved surface. Asmall air bubble is $6\ \mathrm{cm}$ below the flat surface inside it along the axis. The position of the

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iamge of the air bubble made by the mirror is seen :



(A) 14 cm below flat surface

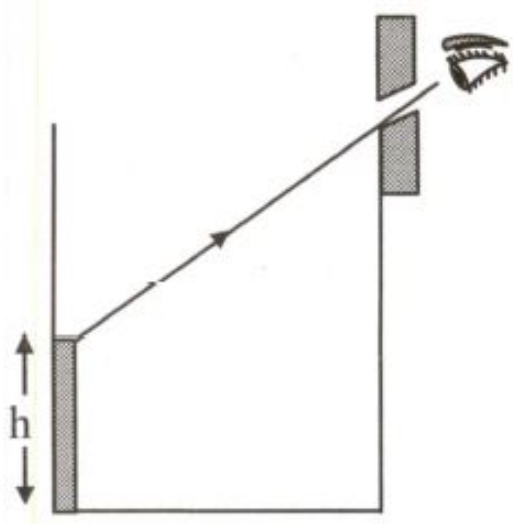
(B) 20 cm below flat surface

(C) 16 cm below flat surface

(D) 30 cm below flat surface

Q.19

A rod of height h is placed in a beaker of same radius. The height of the beaker is 3 times its radius. An observer sees the top end of the rod through a pin-hole (see figure). When the beaker is filled with a liquid of refractive index $\sqrt{\mu}$ upto a height 2h, he can see the lower end of the rod. Find the value of μ.



(A) 2.2

(B) 2.6

(C) 2.9

(D) 2.5

Q.20

Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror and parallel to the second is reflected from the second mirror parallel to the first mirror. The angle between the two mirrors is :-

(A) 30°

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(B) 45°

(C) 60°

(D) 75°

Answers & Solutions

Q.1 Answer:

1.55

Solution:

In the first case we have $(\because f_m = \infty)$

$\frac{1}{28} = \frac{2}{f_1} + \frac{1}{f_m} = \frac{2}{f_1}$ **(i)**

In the second case we have

$\frac{1}{10} = \frac{2}{f_1} + \frac{1}{f_m}$ **(ii)**

where f_m **is the focal length of the curved silvered surface. Hence** $f_m = \frac{R}{2}$ **where** R **is the radius of curvature of the curved surface.**

Subtracting (i) from (ii) we get

$$\frac{1}{f_m} = \frac{1}{10} - \frac{1}{28}$$

$$\text{or } f_m = \frac{140}{9} \text{ cm. Therefore}$$

$R = 2f_m = \frac{280}{9}$ cm

From (i) we have $f_1 = 28 \times 2 = 56$ cm. **Now**

$\frac{1}{f_1} = (\mu - 1)\frac{1}{R}$

or $\mu - 1 = \frac{R}{f_1} = \frac{280}{9 \times 56} = 0.55$

or $\mu = 1.55$

Q.2 Answer:

$\frac{20}{3}$ cm

Solution:

Refer to Fig.

Due to refraction by the slab the image of O **is formed at** I'**. Let** $x = I'Q$ **be the distance of** I' **from side** BC**. Then**

$x = \frac{15 \text{ cm}}{1.5} = 10$ cm

Image I' **serves as the virtual object for concave mirror forming its image** I **due to reflection. Now**

$u = Pr' = -(10 + 5) = -15$ cm

and $v = PI = -(7 + 5) = -12$ cm

Using these values in the spherical mirror formula,

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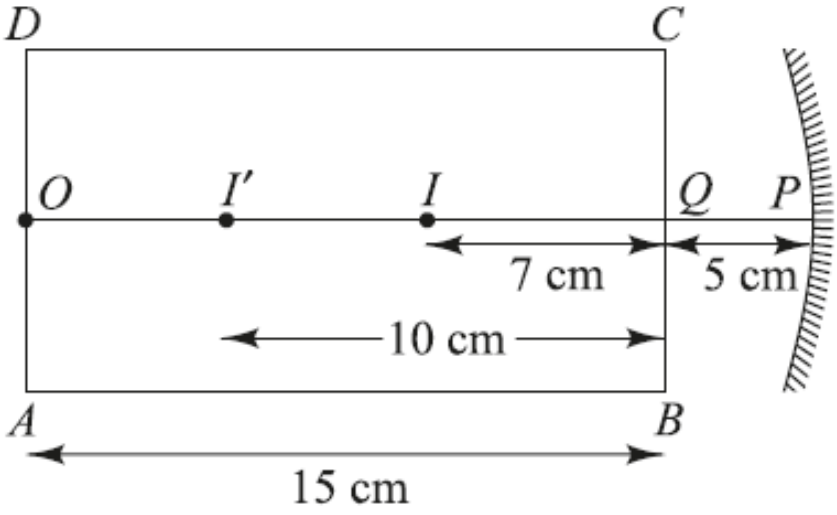
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

we have

$$\frac{1}{f} = \frac{1}{-15} + \frac{1}{-12} = -\frac{3}{20}$$

$$\Rightarrow \quad f = -\frac{20}{3} \text{ cm}$$

So the correct choice is (c).



Q.3 Answer:

$\frac{1}{15}$ m/s

Solution:

From mirror formula

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ so, $\frac{dv}{dt} = -\frac{v^2}{u^2}\left(\frac{du}{dt}\right)$

$\Rightarrow \frac{dv}{dt} = -\left(\frac{f}{u-f}\right)^2 \frac{du}{dt}$

$\Rightarrow \frac{dv}{dt} = \frac{1}{15}$ m/s

Q.4 Answer:

-18.31 cm

Solution:

The image I ' for first refraction (i.e., when the ray comes out of liquid) is at a depth of

$$= \frac{33.25}{1.33} = 25 \text{ cm} \left[\because \text{ Apparent depth } = \frac{\text{Real depth}}{\mu}\right]$$

Now, reflection will occur at concave mirror. For this Γ behaves as an object. $\therefore \quad u =$ $-(15+25) = -40$ cm

and $v = -\left[15 + \frac{25}{1.33}\right]$

Where $\frac{25}{1.33}$ is the real depth of the image.

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Using mirror formula we get

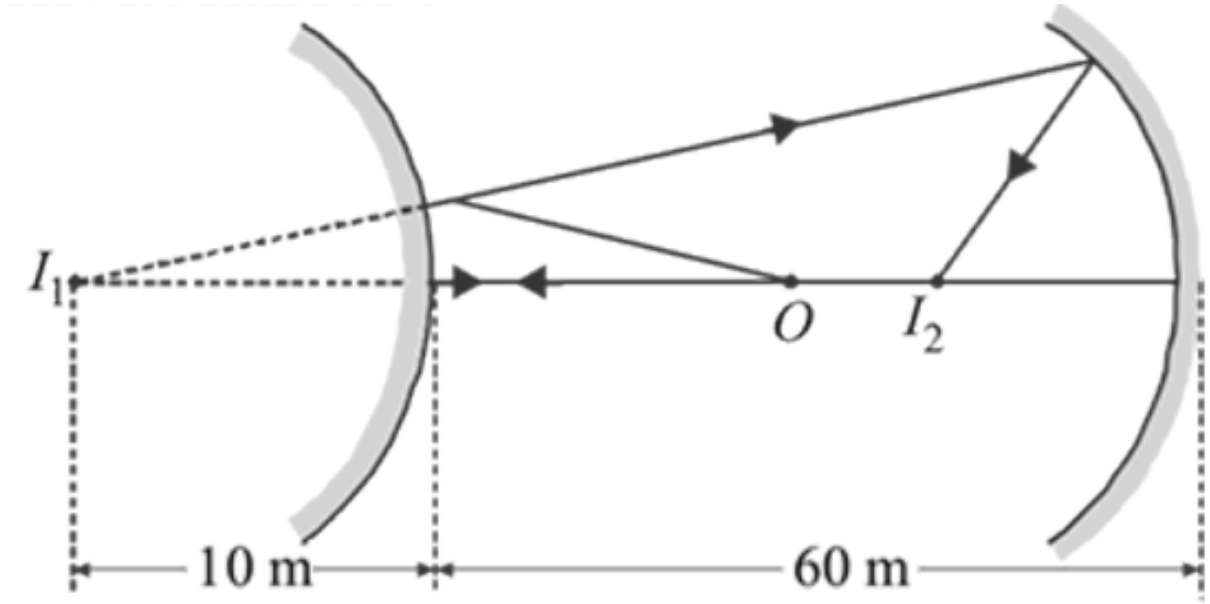
$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}, \quad f=-18.31 \text{ cm}$$

Q.5 Answer:

19.09 cm from the pole of the concave mirror

Solution:

For convex mirror:



$\frac{1}{v}+\frac{1}{-30}=\frac{1}{+15}$ **or** $v=10 \text{ m}$

For concave mirror : $\frac{1}{v}+\frac{1}{-70}=\frac{1}{-15}$

or $v=-19.09 \text{ cm}$

Q.6 Answer:

If O and I are on opposite sides of the principal axis, then they have to be on opposite sides of the mirror as well

Solution:

Use the concepts related to image formation by spherical mirrors.

Q.7 Answer:

$-\left(\frac{R}{R-2U}\right)^2 v_0$

Solution:

For concave mirror $\frac{2}{R}=\frac{1}{v}+\frac{1}{u}$ **or** $\frac{2}{-R}=\frac{1}{v}+\frac{1}{-u}$

$\therefore \quad \frac{1}{v}=\frac{1}{U}-\frac{2}{R}=\frac{R-2U}{UR}$ **or** $v=\left[\frac{RU}{R-2U}\right]$

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In spherical mirror, image velocity $v_i = -\left[\frac{v^2}{u^2}\right] v_0 = -\left[\frac{RU}{R-2U}\right]^2 \frac{v_0}{U^2} = -\left[\frac{R}{R-2U}\right]^2 v_0$

Q.8 Answer:

$R\left(1-\frac{1}{\sqrt{3}}\right)$

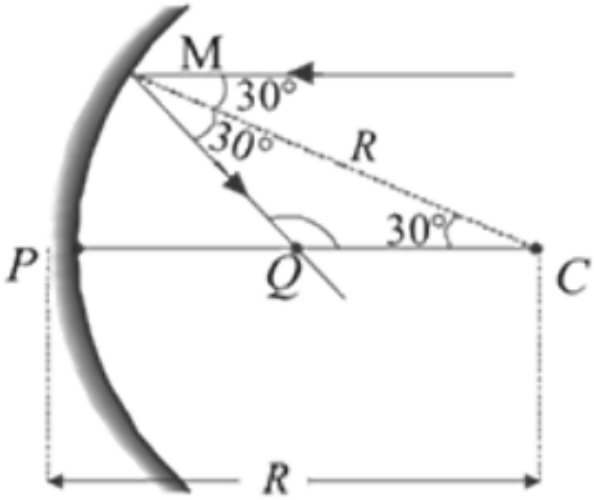
Solution:

From similar triangles, M

$\frac{QC}{\sin 30^\circ} = \frac{R}{\sin 120^\circ}$

or $QC = R \times \frac{\sin 30^\circ}{\sin 120^\circ} = \frac{R}{\sqrt{3}}$

Thus $PQ = PC - QC = R - \frac{R}{\sqrt{3}} = R\left(1-\frac{1}{\sqrt{3}}\right)$



Q.9 Answer:

real, and located at a point between C and O

Solution:

$\frac{1}{\mathrm{V}} + \frac{1}{\mathrm{R}} = \frac{-2}{\mathrm{R}}$

$\mathrm{v} = \frac{-\mathrm{R}}{3}$

this again serves as virtual image for the lens

$\mathrm{u} = \frac{\mathrm{R}}{3}$

$\frac{1}{\mathrm{v}} - \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{f}}$

$\frac{1}{v} - \frac{3}{R} = \frac{2}{R}$

$\mathrm{v} = \frac{\mathrm{R}}{5}$

hence the image is real and formed between O **and** C

Q.10 Answer:

is always less than v

Solution:

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Image speed $= -\left(\frac{v^2}{u^2}\right)\frac{du}{dt}$

For convex mirror $\left|\frac{v}{u}\right| < 1 \therefore$ **Image speed is always less than** v.

Q.11 Answer:

$\ell\left(\frac{f}{u-f}\right)^2$

Solution:

The axial magnification $\delta v = -\frac{v^2}{u^2}(\delta u) = \left(\frac{f}{u-f}\right)^2 \ell$

Q.12 Answer:

$2\mathrm{u}$, in the direction same as that of mirror's velocity

Solution:

Image velocity (w.r.t mirror) $= -\mathrm{m}\times$ **object velocity** (**w.r.t mirror**) **Here** $\mathrm{m} = 1$

Q.13 Answer:

$-60\ \mathrm{cm}$

Solution:

The mirror formula is given as $\frac{1}{\mathrm{v}} + \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{f}}$ **... (i)**

The position of object is $10\ \mathrm{cm}$ **left of mirror and position of image is** $25 - 10 = 15\ \mathrm{cm}$ **right of mirror**

$\mathrm{u} = -10\ \mathrm{cm}$ **and** $\mathrm{v} = 15\ \mathrm{cm}$ **substituting values in equation (i)**

$\frac{1}{15} + \frac{1}{-10} = \frac{1}{f}$

$\frac{1}{\mathrm{f}} = \frac{10-15}{150} \Rightarrow \mathrm{f} = -30\ \mathrm{cm}$

The radius of curvature is given as $\mathrm{R} = 2\mathrm{f} = -60\ \mathrm{cm}$

Q.14 Answer:

$0.25\ \mathrm{m}$

Solution:

For P: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

or $\frac{1}{v_P} + \frac{1}{-3} = \frac{1}{-1}$ **or** $v_P = -\frac{3}{2}m$

For Q : $\frac{1}{v_Q} + \frac{1}{-5} = \frac{1}{-1}$ **or** $v_Q = \frac{-5}{4}m$

So horizontal distance between image of P **and** Q **is** $0.25\ \mathrm{m}$

Q.15 Answer:

$+2$

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Solution:

For M_1 : $V = -60,\ m_1 = -2$
For M_1 : $u = +20,\ F = 10$
$\frac{1}{V} = \frac{1}{20} = \frac{1}{10} \Rightarrow V = 20 \quad \therefore \quad M_2 = -\frac{20}{20} = -1$
$\therefore \quad M = m_1 \times m_2 = +2$

Q.16 Answer:

$3d$

Solution:

So, required distance $= d + d + d = 3d$.

Q.17 Answer:

25 cm

Solution:

$\overrightarrow{V}_{im} = \frac{V^2}{u^2}\overrightarrow{V}_{om}$
$0 - \frac{40}{13} = -\left(\frac{-10}{-10+x}\right)^2\left(10 - \frac{40}{13}\right) \quad \frac{-10}{-10+x} = \pm\frac{2}{3}$
$x = 25, -5$ cm
Object is infront of mirror Hence 25 cm.

Q.18 Answer:

20 cm below flat surface

Solution:

Given, radius of hemispherical glass $R = 10$ cm
- Focal length $f = \frac{10}{2} = -5$ cm
$u = (10 - 6) = -4$ cm
By using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-4} = \frac{1}{-5} \Rightarrow v = 20$ cm
Apparent height, $h_a = h_r\frac{\mu_1}{\mu_2} = 30 \times \frac{1}{1.5} = 20$ cm
below flat surface.

Q.19 Answer:

2.5

Solution:

The line of sight of the observer remains constant, making an angle of $45°$ **with the normal.**
$\sin\theta = \frac{h}{\sqrt{h^2 + (2h)^2}}$

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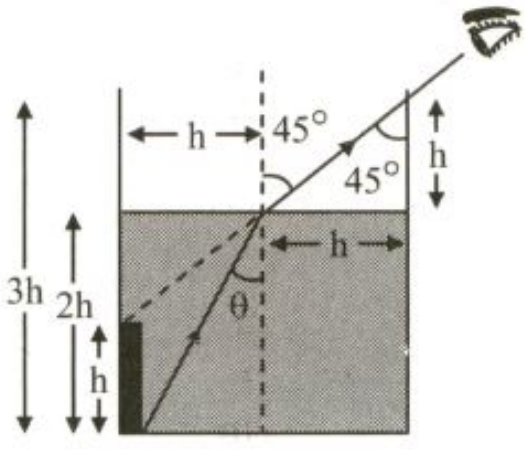
$= \frac{1}{\sqrt{5}}$

Refractive index of liquid,

$\sqrt{\mu} = \frac{\sin 45°}{\sin\theta}$

$= \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{5}}} = \sqrt{\frac{5}{2}}$

$\therefore \quad \sqrt{\mu} = \sqrt{2.5} \quad \Rightarrow \mu = 2.5$



Q.20 Answer:

60°

Solution:

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